

James Hammers

(949) 742-8505 | james.hammers@princeton.edu | [linkedin.com/in/james-hammers](https://www.linkedin.com/in/james-hammers) | jameshamdev.github.io/engineering-portfolio

EDUCATION

Princeton University

Aug. 2024 – May 2028

B.S.E. Electrical & Computer Engineering | Minor: Computer Science | GPA: 3.7

Princeton, NJ

- **Completed Coursework:** Java Programming, Mechanics, Electricity & Magnetism, Linear Algebra, Algorithms and Data Structures, Digital Logic Design, Semiconductor Devices, Multivariable Calculus
- **In Progress and Upcoming:** Systems Programming, Signals, Circuits, Computer Architecture

EXPERIENCE

Electrical Engineering Intern

Incoming Summer 2026

Second Order Effects (SOE) – El Segundo, CA

- Will contribute to end-to-end development of custom electronics across aerospace, automotive, and industrial sectors, including schematic capture, multi-layer PCB layout in Altium Designer, and signal integrity analysis
- Will perform board bring-up, hardware validation, and debug using lab instrumentation (oscilloscopes, logic analyzers, spectrum analyzers), resolving hardware-firmware interactions on client-facing prototypes
- Will gain exposure to SOE's technical stack spanning power electronics, mixed-signal circuits, analog sensor interfaces, motor drivers, and embedded firmware across simultaneous client projects

Undergraduate Researcher – Healthcare AI Laboratory

Jan. 2025 – Present

Penn Medicine Medical Center, under Dr. Jonathan Woo (Chair, Dept. of Medicine)

- Built a low-latency cardiology AI assistant using Flask, FAISS, and LLaMA 3 (via Ollama), integrating RAG-based semantic search over the MIMIC-III clinical dataset to deliver real-time diagnoses and treatment recommendations
- Developed multi-agent systems modeling specialized clinical roles — an autonomous cardiologist agent and collaborative “doctor team” frameworks — to coordinate diagnosis and treatment planning across patient cases
- Applying reinforcement learning under Dr. Jonathan Woo (Chair, Dept. of Medicine) to optimize the clinical decision pipeline, building agentic architectures and refining diagnostic reward signal design

High Voltage and Accumulator Engineering Team Member

Aug. 2024 – Present

Princeton Racing Electric (FSAE)

- Designed custom flexible PCBs in Altium Designer for real-time battery cell monitoring, iterating through full prototyping and test cycles — including panelization, bring-up, validation, and design revision
- Fabricated a custom manufacturing jig to precisely attach flex PCBs to battery cells within the high-voltage pack
- Developed main board wiring and complete Altium schematics for the power distribution system, ensuring compliance with FSAE high-voltage safety requirements

High Power Rocketry and Avionics Team Member

Aug. 2024 – Feb 2026

Princeton Rocketry

- Designed and CAD-modeled a pan-and-tilt Yagi antenna tracker in Onshape, engineering 3D-printed mechanical components for real-time telemetry tracking of active rocket flights
- Built and launched Zephyr Rockets, integrating Arduino-based telemetry systems for real-time data collection

PROJECTS

PUnC — 16-Bit Microprocessor | Verilog, LC-3 ISA, FPGA

Nov. 2025 – Dec. 2025

- Designed and implemented a 16-bit microprocessor in Verilog covering a substantial LC-3 ISA subset, building full RTL including datapath, controller FSM, ALU, PC logic, condition codes, and register-file integration
- Validated design via automated test suite, waveform debugging, and an original LC-3 assembly program; synthesized and deployed to FPGA, confirming end-to-end CPU functionality

Simon Memory Game Machine | Verilog HDL, RTL Design

Nov. 2025

- Implemented a two-player Simon game in Verilog with custom datapath, control FSM, 64-entry pattern memory, comparators, legality-checking logic, counters, and LED display drivers
- Wrote modular testbenches for datapath, controller, and full gameplay; design passed all verification tests

TECHNICAL SKILLS

Languages: Python (incl. Flask), C++ (incl. Embedded), Java, Swift, JavaScript (ES6+)

Frameworks & Libraries: Flask, FAISS, LLaMA 3, React, Firebase

Hardware & Embedded: PCB Design (Altium Designer, KiCad), Flex PCB Design, Microcontrollers (ESP32, Arduino), Embedded C/C++, Sensor Interfacing (I2C), Soldering, Onshape, Siemens NX

HDL & Digital Design: Verilog, RTL Design, FSM Design, FPGA Synthesis & Deployment, LC-3 ISA

Tools & Platforms: Git, GitHub, VS Code, Ollama, Lab Instrumentation (Oscilloscopes, Logic Analyzers, Spectrum Analyzers, and Waveform Analysis)

Concepts: Data Structures & Algorithms, Retrieval-Augmented Generation (RAG), Reinforcement Learning, Embedded Systems, Circuit Design & Analysis